

Relations

Dev - FastAPI

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Relations

What is?

- The way to connect a table with other.
- Allows association between registers.
- Common ways of relationships:
 - One to one (1:1)
 - One to many (1:N)
 - Many to many (N:M)

Relations

How to?

- **1:1** Unique Foreign Key.
- **1:N** Foreign key in the many table.
- **N:M** Is necessary an intermediate table.
 - **Single:** A pivot or link table.
 - **Associative entity:** More attributes.

```
from sqlalchemy import SQLModel, Field, Relationship
```

```
class CustomerBase(SQLModel): 3 usages
```

```
    name: str | None = Field(default=None, nullable=True)
```

```
    description: str | None = Field(default=None, nullable=True)
```

```
class Customer(CustomerBase, table=True): 13 usages
```

```
    id: int | None = Field(default=None, primary_key=True)
```

```
    transactions : list["Transaction"] = Relationship(back_populates="customer")
```

```
class TransactionBase(SQLModel): 2 usages
```

```
    ammount: int | None = Field(default=None, nullable=True)
```

```
    description: str | None = Field(default=None, nullable=True)
```

```
class Transaction(TransactionBase, table=True): 6 usages
```

```
    id: int | None = Field(default=None, primary_key=True)
```

```
    customer_id: int = Field(foreign_key="customer.id" )
```

```
    customer:Customer = Relationship(back_populates="transactions")
```

```
class TransactionCreate(TransactionBase): 2 usages
```

```
    customer_id: int = Field(foreign_key="customer.id" )
```

Customer has a list
of transactions.
Doesn't a field, is a relationship

Transaction has a relation
with customer

When a transaction is created,
need a customer

1:N Relationship

● ● ● models.py

```
class ProductBase(SQLModel): 2 usages
```

```
    name: str
```

```
    price: float
```

```
class Product(ProductBase, table=True): 1 usage
```

```
    id: int | None = Field(primary_key=True, default=None)
```

```
    transactions: list["Transaction"] = Relationship(back_populates="product", link_model=ProductBase)
```

```
class TransactionProductLink(SQLModel, table=True): 1 usage
```

```
    transaction_id: int = Field(primary_key=True, foreign_key="transaction.id")
```

```
    product_id: int = Field(foreign_key="product.id", primary_key=True)
```

```
    quantity: int | None = Field(default=1)
```

Table that link a product with transaction (and quantity)

N:M Relationship

Adding product model and the link table

models.py

```
class Transaction(TransactionBase, table=True): 7 usages
    id: int | None = Field(default=None, primary_key=True)
    customer_id: int = Field(foreign_key="customer.id" )
    customer: Customer = Relationship(back_populates="transactions")

    products: list["Product"] = Relationship(back_populates="transaction", link_model=TransactionProductLink)
```

A list of products in a transaction.

N:M Relationship

Adding products as a list in transaction table

Challenge

- Make a F.U.C.K endpoints for the relationships for practice.
- Research about migrations with alembic

Another examples

- Go to documentation.
- SQLModel Many to many - intro